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import pandas as pd
data = pd.read_excel("games.csv.xlsx")
print("=" * 100)
print("
            FIND-S ALGORITHM")
print("
            MOST SPECIFIC HYPOTHESIS")
print("=" * 100)
print("\nDATASET")
print("-" * 100)
display(data)
attributes = list(data.columns[:-1])
target = data.columns[-1]
print("\nAttributes:")
print(attributes)
print("\nTarget Attribute:")
print(target)
hypothesis = ["Ø"] * len(attributes)
print("\n")
print("=" * 100)
print("INITIAL HYPOTHESIS")
print("=" * 100)
print("H0 =", tuple(hypothesis))
history = []
for index, row in data.iterrows():
    example_no = index + 1
    values = row.iloc[:-1].tolist()
    class_value = str(row.iloc[-1]).strip()
    print("\n")
    print("#" * 100)
    print(f"
            TRAINING EXAMPLE {example_no}")
    print("#" * 100)

    print("\nExample:")
    for attribute, value in zip(attributes, values):
        print(f"{attribute:<30} : {value}")

    print(f"{target:<30} : {class_value}")

    print("\nCurrent Hypothesis BEFORE processing:")
    print(tuple(hypothesis))
    if class_value.lower() == "yes":

        print("\nCLASS = YES")
        print("POSITIVE EXAMPLE")
        print("Processing this example...")

```

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print("\n\n")
print("=" * 100)
print("
                                COMPLETE HYPOTHESIS HISTORY")
print("=" * 100)

```

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history_columns = ["Example", "Class"] + attributes

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history_df = pd.DataFrame(
    history,
    columns=history_columns
)

```

COMPLETE HYPOTHESIS HISTORY

	Example	Class	Sky	AirTemp	Humidity	Wind	Water	Forecast
0	1	Yes	Sunny	Warm	Normal	Strong	Warm	Same
1	2	Yes	Sunny	Warm	?	Strong	Warm	Same
2	3	No	Sunny	Warm	?	Strong	Warm	Same
3	4	Yes	Sunny	Warm	?	Strong	?	?

FINAL RESULT

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Total Training Examples : 4
Total Attributes : 6

Final Most Specific Hypothesis:

Sky	= Sunny
AirTemp	= Warm
Humidity	= ?
Wind	= Strong
Water	= ?
Forecast	= ?

FINAL HYPOTHESIS:

H = ('Sunny', 'Warm', '?', 'Strong', '?', '?')